Inference I: t-tests, F-tests, and Confidence Intervals

Applied Linear Models I • Lecture 4 of 16

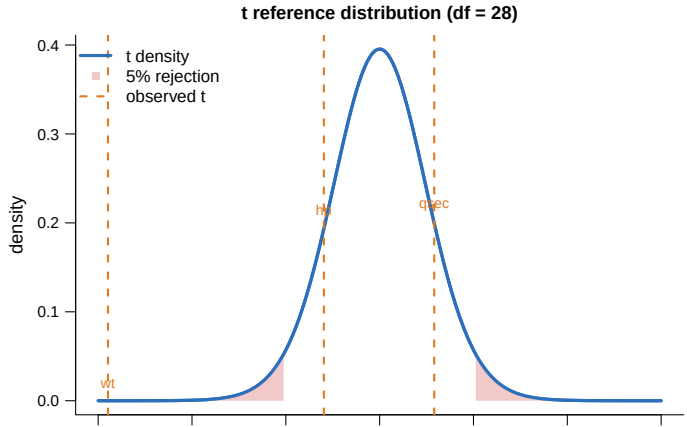
Today

Under normal errors $\varepsilon \sim N(0, \sigma^2 \mathrm{I})$ (exact) or large n (approx):

- t-test for one coefficient.
- overall F-test; partial (nested) F-test; the $F = t^2$ identity.
- general linear hypothesis $\mathrm{L}\boldsymbol{\beta} = \mathrm{c}$.
- confidence intervals and **test–CI duality**.

t-test for a single coefficient

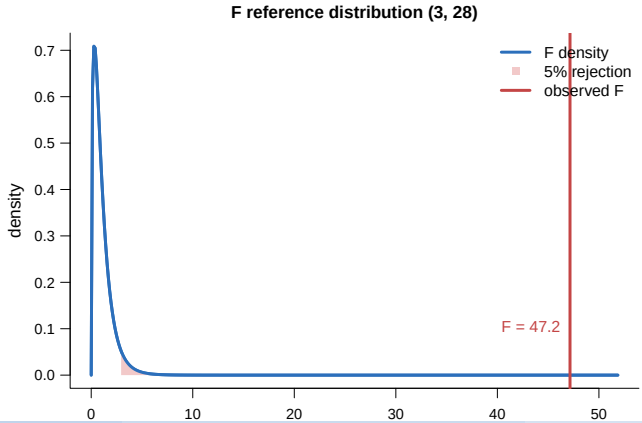
$$t_j = \frac{\hat{\beta}_j - \beta_j^0}{\widehat{\mathrm{SE}}(\hat{\beta}_j)} \sim t_{n-p} \quad \text{under } H_0 : \beta_j = \beta_j^0.$$



Overall F-test

$H_0 : \beta_1 = \cdots = \beta_{p-1} = 0$ vs. at least one nonzero:

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} \sim F_{p-1,\, n-p}.$$



Partial F-test and $F = t^2$

Reduced (nested) vs. full model:

$$F = \frac{(\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}})/q}{\text{SSE}_{\text{full}}/(n-p)} \sim F_{q,\, n-p} \qquad \texttt{anova(reduced, full).}$$

Does qsec add to `wt + hp`?

Partial $F = 1.353$ on $(1, 28)$; and $t^2_{\texttt{qsec}} = 1.163^2 = 1.353$.

For a **single restriction**, the F-test *is* the t-test squared.

General linear hypothesis $\mathsf{L}\boldsymbol{\beta} = \mathsf{c}$

$$F = \frac{(\mathsf{L}\hat{\boldsymbol{\beta}} - \mathsf{c})^\top \left[\mathsf{L}(\mathsf{X}^\top\mathsf{X})^{-1}\mathsf{L}^\top\right]^{-1}(\mathsf{L}\hat{\boldsymbol{\beta}} - \mathsf{c})}{q\,\hat{\sigma}^2} \sim F_{q,\,n-p}.$$

One framework for all tests: pick L.

$H_0 : \beta_{\texttt{wt}} = \beta_{\texttt{qsec}}$, i.e. $\mathsf{L} = (0, 1, 0, -1)$

Quadratic form: $F = 21.12$.

Reparameterized nested model `mpg ~ I(wt+qsec) + hp`: $F = 21.12$.

Identical – two routes to the same test.

Confidence intervals and duality

$$\hat{\beta}_j \pm t_{n-p,\,1-\alpha/2}\,\widehat{\mathrm{SE}}(\hat{\beta}_j).$$

Test–CI duality

The two-sided level-α test of $\beta_j = \beta_j^0$ rejects **iff** β_j^0 is outside the $100(1-\alpha)\%$ CI.

Worked check: "CI excludes 0" agrees with "$p < 0.05$" for every coefficient.

A CI reports the whole range compatible with the data – keeping attention on **effect size**, not just significance.

Summary

- t: one coefficient, partial. Overall F: any slope nonzero.
- Partial F: a subset; for one restriction $F = t^2$.
- $L\boldsymbol{\beta} = c$ unifies them; equivalent to nested-model comparison.
- CIs are dual to tests and report effect sizes – prefer them.

Next

Lecture 5: inference *without* normality – the bootstrap – and prediction intervals.